

Ankit Shukla

Pune, India | 7972628423 | asankitshukla769@gmail.com | <https://www.linkedin.com/in/ankit-shukla-47b8a434/>

PROFESSIONAL SUMMARY

Embedded and systems software engineer with 15 years of C/C++ experience across Embedded Linux, QNX, LynxOS, Windows, automotive, railway, and avionics platforms. Strong background in embedded software, RTOS, hardware-adjacent development, device drivers, interrupts, MMIO, synchronization, IPC, and debugging. Experienced in safety-critical railway and DO-178B avionics software and technical leadership.

TECHNICAL SKILLS

Programming: C, C++11/14/17, Go (Golang), Python, Lua

Scripting & Automation: Linux Shell Scripting, PowerShell Scripting

C++ & Software Design: OOP, STL, SOLID Principles, Design Patterns

Operating Systems & Linux: Linux, POSIX, Process Management, Memory Management, Virtual Memory, VFS, IPC, Multithreading, Concurrency, Synchronization, Deadlocks, NUMA, CPU Scheduling, System Calls

Linux Kernel & Low-Level: Linux Kernel, Device Drivers, Interrupts, MMIO, Hardware Registers, Buffer Queues

Networking & Security: TCP/IP, Network Kernel Stack, OSI Model, Subnetting, HTTP/HTTPS, HTTP/2, SSH, OAuth 2.0, JWT, Load Balancing, Reverse Proxy, API Gateway, Rate Limiting, Internet Gateway (IGW), NAT, nmap

Embedded/RTOS: Embedded Linux, QNX, LynxOS, WinCE, ARM, x86

Automotive: Infotainment, CAN, Voice Commands, Touch/UI

Safety-Critical: DO-178B, ARINC653, ARINC429, DOORS (Requirements Management), Railway Control Systems, Avionics

GPU/Graphics: OpenGL, Mesa, LLVM Graphics Pipeline, libdrm, Linux DRM/KMS, Command Submission

Multimedia: GStreamer, FFmpeg, Audio/Video Codecs, Streaming Pipelines

Distributed Systems & Architecture: Microservices, Monolithic Architecture, Event-Driven Architecture, Control Plane, gRPC, REST APIs, SOAP, WebSocket

Messaging & Streaming: Apache Kafka, RabbitMQ

Cloud & Containerization: Docker, Kubernetes, AWS (EC2, IAM), Azure (VMs, VNETs, ARM Templates)

Databases: SQL, NoSQL

IDEs & Development Tools: QNX Momentics IDE, Eclipse, Visual Studio, Visual Studio Code, Qt Creator, Make, CMake, Bazel, Jenkins, Git

Testing & Debugging: GDB, Trace32/Lauterbach, GoogleTest/GoogleMock, VectorCAST

PROFESSIONAL EXPERIENCE

COHESITY INC. Pune — Data Security, Protection & Management

Software Engineer 4 – Resource Manager (Heimdall)

Dec 2024 – Present

- Understood the requirements of 4 newly assigned features per release and prepared the HLD and LLD.
- Discussed the HLD and LLD docs with 12 stakeholders and feature owners across 4 teams.
- Built new features in C++, Go, and Python, created functional tests, and delivered them across 6 releases.
- Created/updated automation tests and ran them through Ginkgo to ensure no regression is introduced and the new feature works as expected.
- Resolved 10+ release blocker issues across workflows including Linux/Windows VMs and disks.
- Owned the Jenkins CI/CD dev environment and fixed the reported issues in Python scripts used across 4 workflows.

VERITAS TECHNOLOGIES Pune — Data Protection & Backup Software

Principal Software Engineer

Jan 2023 – Dec 2024

- Understood the requirements of 3 newly assigned features per release and prepared/updated design docs.
- Engineered new features in C++ requested by product owners for shared licensing and entitlement validation.
- Resolved 12 reported bugs across the C++ SDK and Java SDK and delivered fixes.
- Supported the licensing component across Windows, Linux, AIX, and Solaris platforms.
- Analysed and remediated 32 reported vulnerabilities (CVEs) in POCC and OpenSSL libraries and provided updated builds with fixes.
- Managed the Jenkins CI/CD dev and production environment used to share builds.

SIEMENS MOBILITY Pune — Railway / Transportation Safety Systems

Senior Software Engineer / Technical Lead

Mar 2019 – Jan 2023

- Led six engineers building safety-critical C/C++ software for railway Radio Block Center (RBC) systems.
- Owned requirements, design, implementation, static analysis, testing, code reviews, release, and mentoring across 12 releases of the RBC system.
- Implemented Python automation tools to streamline the process, reducing manual effort by ~40% per release across the team.
- Created 350+ unit tests in VectorCAST to measure code coverage, achieving 98% coverage.

TATA ELXSI Pune — Automotive Embedded Systems

Specialist — Automotive Embedded Systems

Jun 2015 – Mar 2019

- Designed and built C/C++ automotive infotainment connectivity software on QNX and Windows.
- Worked on voice-command handling, touch/UI input, CAN bus integration, debugging, and defect resolution.
- Analysed and resolved 65 reported bugs using log analysis.
- Diagnosed and resolved 16 core dump issues using shared dump files.
- Ran static code coverage analysis to meet internal quality standards, achieving 96% coverage.

INFOSYS Hyderabad — Semiconductor / GPU Graphics

Senior Software Engineer — Graphics Driver Development

Aug 2014 – Jun 2015

- Worked on the AMD Radeon GPU driver stack across OpenGL, Mesa, LLVM, libdrm, and Linux DRM/KMS.
- Debugged reported performance issues using Visual Studio tools (debugger, profiler), improving rendering/frame-rate by 10%.
- Designed a 2D/3D test GUI application using OpenGL ES.

ROCKWELL COLLINS Hyderabad — Avionics / Aerospace

Software Engineer

Aug 2011 – Aug 2014

- Authored DO-178B compliant C/C++ avionics software on LynxOS.
- Worked with ARINC communication protocols and Trace32 debugging.
- Managed and updated ~150 system requirements in DOORS, working with ARINC653 and ARINC429 avionics data bus standards.
- Created 150+ functional tests in Lua to verify system behaviour.

EDUCATION

Diploma in System Software, Pune University	2011
B.Tech – Computer Science & Engineering, Uttar Pradesh Technical University (UPTU)	2010
